Geometría diferencial y aplicaciones

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Escuela Técnica Superior de Ingeniería

Nombre y apellidos:

Teoría: Segunda identidad de Bianchi.

1.- Sea (M,g) una variedad riemanniana. Supongamos que, en una carta (U,φ) de coordenadas (x^1,x^2,x^3,x^4), su campo tensorial métrico es

$$g = c\,dx^1 \otimes dx^1 + \sum_{i,j=2}^{4} f^2(x^1)a_{ij}(x^2,x^3,x^4)dx^i \otimes dx^j$$

donde: $f \in \mathcal{F}(U)$ depende solo de x^1 y $f>0$, $a_{ij} \in \mathcal{F}(U)$ no dependen de x^1 y c es constante. Sean dos campos $\left\{X = \sum_{i=1}^{4} \xi^i \frac{\partial}{\partial x^i}, Y = \sum_{i=1}^{4} \eta^i \frac{\partial}{\partial x^i}\right\} \subset \mathfrak{X}(U)$ tales que: $\xi^1 = \eta^1 = 0$ y las componentes $\xi^i(x^2,x^3,x^4)$, $\eta^i(x^2,x^3,x^4)$ no dependen de x^1. Demostrar que $R(X,Y)\frac{\partial}{\partial x^1} = 0$; para ello, demostrar previamente que $\nabla_X \frac{\partial}{\partial x^1} = \frac{f'}{f}X$, donde f' es la derivada de f respecto de su variable x^1.

2.- Sean $\eta = \sum_{i=1}^{3} \eta_i \vec{e}^{\,*}_i$ y $\theta = \sum_{i=1}^{3} \theta_i \vec{e}^{\,*}_i$, dos 1-tensores de $\Omega^1(E)$, con $\{\vec{e}_i\}_{i=1}^{i=3}$ base ortonormal de un espacio vectorial E, y $\{\vec{e}^{\,*}_i\}_{i=1}^{i=3}$ su base dual. Sean los vectores $\eta^* = \sum_{i=1}^{3} \eta_i \vec{e}_i$, $\theta^* = \sum_{i=1}^{3} \theta_i \vec{e}_i$. Sea $\vec{x} = \sum_{i=1}^{3} x_i \vec{e}_i$ un vector del subespacio $\langle \eta^*, \theta^* \rangle \subset E$. Demostrar que existe $\vec{y} = \sum_{i=1}^{3} y_i \vec{e}_i \in \langle \eta^*, \theta^* \rangle$, otro vector tal que $\eta \wedge \theta = \vec{x}^* \wedge \vec{y}^*$, con $\vec{x}^* = \sum_{i=1}^{3} x_i \vec{e}^{\,*}_i$ y $\vec{y}^* = \sum_{i=1}^{3} y_i \vec{e}^{\,*}_i$.

3.- Sea $\mathcal{S} \hookrightarrow \mathbb{A}^3$ una superficie en el afín euclídeo. Sean Ric y g, sus campos tensoriales de Ricci y métrico inducido, respectivamente. Sea la parametrización $\vec{x}$ de $\mathcal{S}$:

$$\begin{array}{ccl} (0,\pi)\times(0,2\pi) & \to & \mathbb{R}^3 \\ (u,v) & \mapsto & \vec{x}(u,v) = \left(\sin u \cos v, \sin u \sin v, \sqrt{2}\cos u\right) \end{array}$$

Demostrar que $Ric = Kg$ con K su campo escalar de curvatura de Gauss.

① $R[X,Y]\frac{\partial}{\partial x^1} = \nabla_X\nabla_Y \frac{\partial}{\partial x^1} - \nabla_Y\nabla_X\frac{\partial}{\partial x^1} - \nabla_{[X,Y]}\frac{\partial}{\partial x^1} \overset{*}{\underset{A}{=}}$

$$= \nabla_X\left(\frac{f'}{f}Y\right) - \nabla_Y\left(\frac{f'}{f}X\right) - \frac{f'}{f}[X,Y] = \frac{f'}{f}\left(\nabla_X Y - \nabla_Y X - [X,Y]\right) = 0$$

Esto es porque f solo depende de x^1, y los campos X, Y no tienen componente de $\frac{\partial}{\partial x^1}$

Ⓐ Se aplica que $\nabla_X \frac{\partial}{\partial x^1} = \frac{f'}{f}X$, pero también paso con Y y con $[X,Y]$ ya que ambos campos son de la forma de X, o sea no tienen componente de $\frac{\partial}{\partial x^1}$, y las demás componentes no dependen de x^1.

* Así que hay que demostrar que $\nabla_X \frac{\partial}{\partial x^1} = \frac{f'}{f}X$:

Basta demostrar que $g\left(\nabla_X \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^i}\right) = g\left(\frac{f'}{f}X, \frac{\partial}{\partial x^i}\right)\ \forall \frac{\partial}{\partial x^i}$..

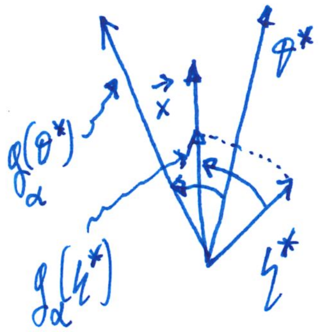
$$\begin{cases} g\left(\frac{f'}{f}X,\frac{\partial}{\partial x^1}\right) = \frac{f'}{f}c\,\xi^1 = 0 \\ g\left(\nabla_X \frac{\partial}{\partial x^1},\frac{\partial}{\partial x^1}\right) = X g\left(\frac{\partial}{\partial x^1},\frac{\partial}{\partial x^1}\right) - g\left(\frac{\partial}{\partial x^1},\nabla_X\frac{\partial}{\partial x^1}\right) \Rightarrow g\left(\nabla_X \frac{\partial}{\partial x^1},\frac{\partial}{\partial x^1}\right) = \\ = \frac{1}{2}X(c) = 0 \end{cases}$$

$$k\neq 1\begin{cases} g\left(\frac{f'}{f}X,\frac{\partial}{\partial x^k}\right) \overset{k\neq1}{=} \frac{f'}{f}\sum_{i,j=2} f^2 a_{ij}\, dx^i(X)\, \underbrace{dx^j\left(\frac{\partial}{\partial x^k}\right)}_{\delta^j_k} = \frac{f'}{f}\sum_{i=2} f^2 a_{ik}\,\xi^i = \sum_{i=2} ff'\xi^i a_{ik} \\ g\left(\nabla_X \frac{\partial}{\partial x^1},\frac{\partial}{\partial x^k}\right) = X\underbrace{g\left(\frac{\partial}{\partial x^1},\frac{\partial}{\partial x^k}\right)}_{=0} - g\left(\frac{\partial}{\partial x^1},\nabla_X\frac{\partial}{\partial x^k}\right) \underset{\xi^1=0}{=} -\sum_{i=2}\xi^i g\left(\frac{\partial}{\partial x^1},\nabla_{\frac{\partial}{\partial x^i}}\frac{\partial}{\partial x^k}\right) = \\ = -\sum_{i=2}\xi^i g\left(\frac{\partial}{\partial x^1},\sum_r \Gamma^r_{ik}\frac{\partial}{\partial x^r}\right) = -\sum_i \xi^i c\,\Gamma^1_{ik} = -\sum_{i=2}\xi^i \cancel{c}\,\frac{-1}{\cancel{c}} ff' a_{ik} \quad \underline{\text{q.e.d.}} \\ \Gamma^1_{ik} = \frac{1}{2}\sum_\alpha g^{\alpha1}\left(\frac{\partial g_{k\alpha}}{\partial x^i} + \frac{\partial g_{i\alpha}}{\partial x^k} - \frac{\partial g_{ik}}{\partial x^\alpha}\right) = \frac{1}{2}\frac{1}{c}\left(\frac{\partial g_{k1}}{\partial x^i}+\frac{\partial g_{i1}}{\partial x^k} - \frac{\partial g_{ik}}{\partial x^1}\right) = -\frac{1}{2}\frac{1}{c} ff' a_{ik} \end{cases}$$

② El problema 6 de la lista de problemas de álgebra tensorial enuncia que:

⊛ $\omega \in \Omega^2(E)$ con $\omega = \eta \wedge \vartheta$, $\eta = \sum_i \eta_i e_i^*$, $\vartheta = \sum_i \vartheta_i e_i^*$, $\{\vec{e}_i\}$ base ortonormal del espacio E, $\eta^* = \sum_i \eta_i \vec{e}_i$, $\vartheta^* = \sum_i \vartheta_i \vec{e}_i$. Entonces $\omega(\vec{v}, \vec{w})$ consiste en el área orientada, del paralelogramo P, proyectado sobre el plano vectorial generado por los vectores $\langle \eta^*, \vartheta^* \rangle$, donde ese P está generado por $\vec{v}$ y $\vec{w}$; y luego multiplicar ese área por el área del paralelogramo generado por η^* y ϑ^*.
La dimensión de E es 3, y la proyección es la ortogonal.

Entonces:



Sea $g_\alpha(\eta^*)$ el vector girado en $< \eta^*, \vartheta^* >$ tal que $\frac{1}{\lambda} g_\alpha(\eta^*) = \vec{x}$ con $\lambda > 0$. Así $\vec{y} = \lambda g_\alpha(\vartheta^*)$, el girado de ϑ^* por λ es el vector pedido.

Esto es cierto porque

$$A(\eta^*, \vartheta^*) = A(g_\alpha(\eta^*), g_\alpha(\vartheta^*)) = A\left(\frac{g_\alpha(\eta^*)}{\lambda}, g_\alpha(\vartheta^*) \cdot \lambda\right) =$$

área del paralelogramo generado por η^* y ϑ^*

$$= A(\vec{x}, \vec{y})$$

$\Downarrow \leftarrow$ ⊛

$$\eta \wedge \vartheta(\vec{v}, \vec{w}) = \vec{x}^* \wedge \vec{y}^*(\vec{v}, \vec{w})$$

q.e.d.

③ Por supuesto que puede hacerse con un cálculo largo directo con $\vec{x}$, pero realmente es un teorema general de cualquier superficie $S \hookrightarrow \mathbb{A}^3$; veámoslo:

Sea $\{\vec{e}_i\}$ una base ortonormal de $T_p(S)$

$$\begin{cases} Ric(\vec{e}_i,\vec{e}_j) = \sum_{k=1}^{2} g(R(\vec{e}_k,\vec{e}_i)\vec{e}_j,\vec{e}_k) = \begin{cases} 0 & \text{si } i\neq j \Leftrightarrow \begin{cases} i=1, j=2 \\ i=2, j=1 \end{cases} \\ g(R(\vec{e}_a,\vec{e}_i)\vec{e}_i,\vec{e}_a) & \text{si } i=j \text{ con } a\neq i=j \end{cases} \\ K\, g(\vec{e}_i,\vec{e}_j) = \begin{cases} 0 & \text{si } i\neq j \\ K\cdot 1 = K & \text{si } i=j \end{cases} \end{cases}$$

$g(R(\vec{e}_a,\vec{e}_i)\vec{e}_i,\vec{e}_a) = K$ curvatura de Gauss de S en p.

$$\Rightarrow Ric(\vec{e}_i,\vec{e}_j) \underset{(*)}{=} K\, g(\vec{e}_i,\vec{e}_j)$$

Ahora:

$$Ric(\bar{X},Y)_p = Ric\left(\sum_{m=1}^{2} \xi^m \vec{e}_m, \sum_{n=1}^{2} \eta^n \vec{e}_n\right) =$$

$$= \sum_{m,n=1}^{2} \xi^m \eta^n Ric(\vec{e}_m,\vec{e}_n) \underset{(*)}{=} \sum_{m,n=1}^{2} \xi^m \eta^n K\, g(\vec{e}_m,\vec{e}_n) =$$

$$= K\, g\left(\sum_{m=1}^{2} \xi^m \vec{e}_m, \sum_{n=1}^{2} \eta^n \vec{e}_n\right) = K\, g(\bar{X},Y)\Big|_p \quad \forall p\in S$$

q.e.d.

$\boxed{\text{Anexo}}$

(*) Por si alguien quiere ver una demostración de cómputo directo largo del problema 3:

Paso 1: Calculo las métricas g y g^{-1} de S con $x^1=u$, $x^2=v$.

$$\left\langle \frac{\partial\vec{x}}{\partial u}, \frac{\partial\vec{x}}{\partial u} \right\rangle = E = g_{11} = 1+\operatorname{sen}^2 u, \quad \left\langle \frac{\partial\vec{x}}{\partial u}, \frac{\partial\vec{x}}{\partial v} \right\rangle = F = g_{12} = g_{21} = 0$$

$$\left\langle \frac{\partial\vec{x}}{\partial v}, \frac{\partial\vec{x}}{\partial v} \right\rangle = G = g_{22} = \operatorname{sen}^2 u \Rightarrow g^{11} = \frac{1}{1+\operatorname{sen}^2 u}, \; g^{22} = \frac{1}{\operatorname{sen}^2 u}, \; g^{21} = g^{12} = 0$$

Paso 2: Calculo las Γ_{ij}^{k} de la conexión de Levi-Civita.

$$\Gamma_{11}^{1} = \frac{\cos u \operatorname{sen} u}{1+\operatorname{sen}^2 u}, \; \Gamma_{11}^{2} = 0, \; \Gamma_{22}^{1} = -\frac{\cos u \operatorname{sen} u}{1+\operatorname{sen}^2 u}, \; \Gamma_{22}^{2} = 0, \; \Gamma_{12}^{1} = \Gamma_{21}^{1} = 0$$

$$\Gamma_{12}^{2} = \Gamma_{21}^{2} = \frac{\cos u}{\operatorname{sen} u}.$$ Pongo $\Delta = \det(g_{ij}) = \operatorname{sen}^2 u (1+\operatorname{sen}^2 u)$

Hay que ver que $R_{ij} = K g_{ij}$.

Paso 3 Desarrollo $K g_{ij}$:

$$K g_{ij} = g_{ij} \frac{R_{1221}}{\Delta} = \frac{g_{ij}}{\Delta} g\left(R\left(\frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}\right)\frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^1}\right) = \frac{g_{ij}}{\Delta} g\left(\sum_{m=1}^{2} R_{122}^{m} \frac{\partial}{\partial x^m}, \frac{\partial}{\partial x^1}\right) =$$

$$\underset{g_{12}=0}{=} \frac{g_{ij}}{\Delta} R_{122}^{1} g_{11}$$

Paso 4 Calculo R_{122}^{1}.

$$R_{122}^{1} = \frac{\partial \Gamma_{22}^{1}}{\partial x^1} - \frac{\partial \Gamma_{12}^{1}}{\partial x^2} + \sum_{\alpha} \left(\Gamma_{22}^{\alpha} \Gamma_{1\alpha}^{1} - \Gamma_{12}^{\alpha} \Gamma_{2\alpha}^{1} \right) = 2 \frac{\operatorname{sen}^2 u}{(1+\operatorname{sen}^2 u)^2}$$

Paso 5: Concluyo Kg_{ij}, con lo calculado anteriormente.

$$Kg_{ij} = \begin{cases} 0 & \text{si } i \neq j \\ \dfrac{2}{1+\operatorname{sen}^2 u} & \text{si } i=j=1 \\ \dfrac{2\operatorname{sen}^2 u}{(1+\operatorname{sen}^2 u)^2} & \text{si } i=j=2 \end{cases}$$

Paso 6: Desarrollo R_{ij}.

$$R_{ij} = \sum_{m=1}^{2} R^m_{mij} = \sum_m \left(\frac{\partial \Gamma^m_{ij}}{\partial x^m} - \frac{\partial \Gamma^m_{mj}}{\partial x^i} + \sum_{\alpha=1}^{2} \left(\Gamma^\alpha_{ij}\Gamma^m_{m\alpha} - \Gamma^\alpha_{mj}\Gamma^m_{i\alpha} \right) \right) =$$

$$= \frac{\partial \Gamma^1_{ij}}{\partial x^1} - \frac{\partial \Gamma^1_{1j}}{\partial x^i} + \sum_\alpha \left(\Gamma^\alpha_{ij}\Gamma^1_{1\alpha} - \Gamma^\alpha_{1j}\Gamma^1_{i\alpha} \right) + \frac{\partial \Gamma^2_{ij}}{\partial x^2} - \frac{\partial \Gamma^2_{2j}}{\partial x^i} + \sum_\alpha \left(\Gamma^\alpha_{ij}\Gamma^2_{2\alpha} - \Gamma^\alpha_{2j}\Gamma^2_{i\alpha} \right)$$

Paso 7: Calculo R_{ij}.

$$\left.\begin{array}{l} \underset{i\neq j}{R_{ij}} = 0-0+\dots-0 = 0 \\ R_{11} = -\dfrac{\partial \Gamma^2_{21}}{\partial x^1} + \Gamma^1_{11}\Gamma^2_{21} - \Gamma^2_{12}\Gamma^2_{12} = \dfrac{2}{1+\operatorname{sen}^2 u} \\ R_{22} = \dfrac{\partial \Gamma^1_{22}}{\partial x^1} + \Gamma^1_{22}\Gamma^1_{11} - \Gamma^2_{12}\Gamma^1_{22} = \dfrac{2\operatorname{sen}^2 u}{(1+\operatorname{sen}^2 u)^2} \end{array}\right\} = R_{ij}$$

Así $Ric = Kg$.

q.e.d.

(*) Por si alguien quiere ver otra demostración de cómputo más largo del problema f:

$$\nabla_X \frac{\partial}{\partial x^1} = \nabla_{\sum_{i=2}^{4} \xi^i \frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^1} = \sum_{i=2} \xi^i \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^1} = \sum_{i=2} \xi^i \left(\sum_{k=1}^{4} \Gamma_{i1}^{k} \frac{\partial}{\partial x^k} \right) =$$

$$= \sum_{i=2} \xi^i \left(\sum_{k=1} \left(\frac{1}{2} \sum_{\alpha=1}^{4} g^{\alpha k} \left(\frac{\partial g_{1\alpha}}{\partial x^i} + \frac{\partial g_{i\alpha}}{\partial x^1} - \frac{\partial g_{i1}}{\partial x^\alpha} \right) \right) \frac{\partial}{\partial x^k} \right) = \quad g_{1\beta} = cte$$

$$= \sum_{i=2} \xi^i \left(\sum_{k=1} \left(\frac{1}{2} \sum_{\alpha=2} g^{\alpha k} \frac{\partial g_{i\alpha}}{\partial x^1} \right) \frac{\partial}{\partial x^k} \right) = \sum_{i=2} \xi^i \left(\sum_{k=1} \left(\frac{1}{2} \sum_{\alpha=2} g^{\alpha k} \frac{\partial}{\partial x^1} \left(f^2(x^1)\, a_{i\alpha} \right) \right) \frac{\partial}{\partial x^k} \right) =$$

$$= \sum_{i=2} \xi^i \left(\sum_{k=1} \left(\frac{1}{2} \sum_{\alpha=2} g^{\alpha k}\, 2 f f' a_{i\alpha} \right) \frac{\partial}{\partial x^k} \right) = \sum_{k=1}^{4} \left(\sum_{i,\alpha=2}^{4} \xi^i g^{\alpha k} f f' a_{i\alpha} \right) \frac{\partial}{\partial x^k} =$$

$$= \sum_{k=1} \left(\sum_{i,\alpha=2} \xi^i f f' \frac{1}{f^2} a^{\alpha k} a_{i\alpha} \right) \frac{\partial}{\partial x^k} = \sum_{k=1} \frac{f'}{f} \left(\sum_{i=2} \xi^i \delta_i^k \right) \frac{\partial}{\partial x^k} =$$

$(a^{\alpha k})$ matriz inversa de $(a_{\alpha k})$

$$= \sum_{k=2}^{4} \frac{f'}{f} \xi^k \frac{\partial}{\partial x^k} = \frac{f'}{f} X$$

q.e.d.